Polynomial Links as Parameter-Tied Rank Lifts for Tucker Tensors

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Abstract

Tucker decomposition is a compact model for multiway data, but its map from a latent tensor to measurements is linear. Nonlinear measurement responses can therefore make a low-rank latent tensor appear higher-rank in the measured tensor. We propose Nonlinear Tucker Decomposition with Polynomial Links (NTD-PL), a Tucker enhancement that learns a polynomial link jointly with a Tucker backbone. NTD-PL keeps the backbone ranks fixed and adds only scalar link coefficients. Its polynomial powers induce a parameter-tied rank lift: each degree forms Veronese-lifted Tucker factors tied to the same latent tensor, rather than an untied high-rank correction.

We analyze this structure through Veronese rank expansion, separation from matched-rank Tucker, and approximation results for nonlinear responses. We optimize a masked reconstruction and completion objective by alternating Link Refresh, a small ridge update for the scalar link, with Linked-Backbone Descent and degree continuation. Experiments on controlled tensors, hyperspectral reconstruction and completion, rank-lift and Link Refresh diagnostics, and external cross-domain cases show scoped gains: NTD-PL helps when the residual left by matched-rank Tucker is coherent with a nonlinear response.

1 Introduction

Tensor decompositions provide compact models for multiway data in scientific data analysis and signal processing [Kolda and Bader, 2009, Sidiropoulos et al., 2017, Liu et al., 2013]. Tucker decomposition is a standard choice because it separates a small core from mode-wise factors. It uses a linear map from the Tucker latent tensor to the measured tensor: the reconstructed Tucker value is fitted directly to the measurement. This assumption is natural for many data sets, but it can be mismatched to measurement or data-generation pipelines with response curves, saturation, calibration effects, or material-dependent nonlinearities [Debevec and Malik, 1997, Grossberg and Nayar, 2004, Bioucas-Dias et al., 2012, Heylen et al., 2014]. In such settings, the latent signal may be well organized by a low-rank Tucker tensor while the measured tensor appears to have larger multilinear rank after a nonlinear response.

A direct remedy is to increase Tucker rank, but this changes the multilinear budget and expands the core and factors throughout the model. More flexible nonlinear tensor decoders based on kernels, Gaussian processes, or neural networks can also be useful, but they address a broader problem and often make the relation to a Tucker backbone less explicit [Xu et al., 2011, Zhe et al., 2016, Liu et al., 2017, Saragadam et al., 2024]. We ask a narrower question: can a Tucker model keep its chosen backbone ranks fixed while learning a response that explains coherent nonlinear residuals left by matched-rank Tucker?

We instantiate this idea with a polynomial link. Let

$$\mathcal{S} = [\![\mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)}]\!]$$

be a Tucker latent tensor. Instead of fitting the measured tensor directly by $\mathcal{S}$, we model

$$\widehat{\mathcal{Y}} = f_{\boldsymbol{\beta}}(\mathcal{S}) = \sum_{p=0}^{P} \beta_p \mathcal{S}^{\odot p},$$

where $\odot p$ denotes the Hadamard power. We call the model Nonlinear Tucker Decomposition with Polynomial Links (NTD-PL). It contains Tucker as the degree-one case, keeps the Tucker backbone and multilinear ranks fixed, and adds only $P+1$ scalar link coefficients. The higher powers $\mathcal{S}^{\odot p}$ induce high-order interactions among the same core and factors, so the added expressivity is parameter-tied to one learned latent tensor.

This viewpoint is central to the paper. NTD-PL is best understood as a Tucker enhancement and shared-backbone rank lift, not as a generic nonlinear decoder. A degree-p power of the latent tensor behaves like an ordinary Tucker tensor whose factors are polynomial feature lifts of the original factors, but those lifted features remain tied to one learned backbone. The new degrees therefore reshape the low-rank geometry through the polynomial link, rather than adding an untied high-rank tensor block. This is also why the link is not a post-hoc calibration layer.

Our claims are intentionally scoped. Increasing Tucker rank remains the right comparison when the residual is ordinary multilinear structure, and we do not claim to dominate neural, kernel, or other nonlinear tensor factorization methods. The intended regime is more specific: the residual left by matched-rank Tucker should be coherent with a nonlinear response.

This scoped regime is also supported empirically. In controlled tensors, the gain grows with the injected nonlinear residual energy and with the degree needed to approximate the response. On real hyperspectral tensors, NTD-PL improves matched-rank Tucker in both full reconstruction and held-out completion. More importantly, a one-step Link Refresh diagnostic predicts when NTD-PL will help: across HSI scenes and non-HSI tensor units, larger diagnostic reduction is associated with larger NTD-PL gains, while low-yield units show little or no benefit. These results suggest that nonlinear responses are identifiable structure in several real tensor families.

Contributions. We make the following contributions.

- **Model.** We propose NTD-PL, a Tucker enhancement with a polynomial link on a Tucker latent tensor. It keeps the Tucker backbone and multilinear ranks fixed while adding only scalar coefficients for a parameter-tied response.
- **Theory.** We characterize the induced model class through Veronese rank expansion, separation from matched-rank Tucker, and approximation results for nonlinear responses.
- **Optimization.** We derive a masked objective for reconstruction and completion, and optimize it with Link Refresh, Linked-Backbone Descent, and degree continuation.
- **Experiments.** We evaluate controlled nonlinear residuals, hyperspectral reconstruction and completion, rank-lift and Link Refresh diagnostics, and external cross-domain boundary cases.

2 Related Work

Low-rank tensor models and Tucker backbones. CP, Tucker, tensor train, and tensor ring decompositions provide compact representations for multiway data through different rank notions and factor topologies [Kolda and Bader, 2009, Sidiropoulos et al., 2017, De Lathauwer et al., 2000, Oseledets, 2011, Zhao et al., 2016]. Tucker is the closest baseline for NTD-PL because it uses the same core-factor multilinear backbone. NTD-PL does not primarily compete by changing the rank tuple, replacing the factor topology, or moving to another tensor-network format. Instead, it keeps the Tucker backbone and multilinear ranks fixed and changes the measurement map on top of the Tucker latent tensor.

Nonlinear tensor factorization and neural/kernel decoders. Nonlinear tensor models include generalized likelihood and link formulations, Bayesian and kernelized tensor factorizations, Gaussian-process tensor models, and neural tensor decoders [Hong et al., 2020, Schein et al., 2016, Xu et al., 2011, Zhe et al., 2016, Liu et al., 2017, Pan et al., 2020, Saragadam et al., 2024, Varolgunes et al., 2023]. These methods can be more flexible than a scalar polynomial link, but this flexibility often comes with heavier inference, additional hyperparameters, or a less transparent finite-dimensional relation to Tucker. NTD-PL is complementary: it deliberately uses a low-dimensional scalar response rather than a general kernel, Gaussian-process, or neural decoder, so that the induced effect can be analyzed relative to a fixed Tucker backbone.

Polynomial feature lifts and high-order interactions. Polynomial and Volterra-style expansions are classical tools for representing nonlinear responses [Volterra, 1959, Boyd and Chua, 1985]. Related matrix and tensor methods use polynomial feature maps, quadratic factorizations, or explicit high-order interactions to enlarge a linear model [Zhai et al., 2024, Liu et al., 2015, Deng et al., 2016]. Such constructions can make nonlinear interactions explicit, but untied high-order parameterizations can increase parameter dimension and contraction cost. NTD-PL does not introduce an independent high-order tensor parameter. Its polynomial powers are powers of one learned Tucker latent tensor, inducing parameter-tied high-order interactions and Veronese-style backbone-tied rank lifts.

Positioning. Existing nonlinear tensor models either use broad nonlinear decoders, modify likelihoods or kernels, or explicitly parameterize high-order interactions. What is less developed is a Tucker-preserving, low-dimensional, analyzable polynomial link that yields a parameter-tied rank lift and supports masked learning. NTD-PL trades general nonlinear flexibility for structure, interpretability, and a direct finite-dimensional rank-lift view. Its expected benefit is therefore scoped to settings where matched-rank Tucker leaves a coherent nonlinear residual explained by a scalar response.

3 Model

3.1 Model and Measurement Assumption

Let $\mathcal{Y} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$ be a measured tensor and let

$$\mathcal{S} = [\![\mathcal{X}; \mathbf{A}^{(1)}, \ldots, \mathbf{A}^{(N)}]\!], \qquad \mathcal{X} \in \mathbb{R}^{R_1 \times \cdots \times R_N}, \quad \mathbf{A}^{(n)} \in \mathbb{R}^{I_n \times R_n}. \tag{1}$$

Our measurement assumption is that the measured tensor is generated from this low-rank Tucker latent tensor through a scalar response, up to noise. NTD-PL models this response by a polynomial:

$$Y_{\mathbf{i}} = f_{\boldsymbol{\beta}}(S_{\mathbf{i}}) + \varepsilon_{\mathbf{i}}, \qquad f_{\boldsymbol{\beta}}(s) = \sum_{p=0}^{P} \beta_p s^p, \qquad \mathbf{i} = (i_1, \ldots, i_N). \tag{2}$$

Equivalently, the tensor prediction is

$$\widehat{\mathcal{Y}} = f_{\boldsymbol{\beta}}(\mathcal{S}) = \sum_{p=0}^{P} \beta_p \mathcal{S}^{\odot p}, \qquad \boldsymbol{\beta} \in \mathbb{R}^{P+1}. \tag{3}$$

The constant term β_0 captures affine response shifts, the linear term $\beta_1 \mathcal{S}$ is the usual Tucker channel, and terms with $p \geq 2$ activate nonlinear channels on the same latent tensor. Standard Tucker is recovered by setting $\beta_1 = 1$ and $\beta_p = 0$ for $p \neq 1$. Thus NTD-PL changes the measurement map while leaving the Tucker backbone and its multilinear ranks fixed.

3.2 Backbone-Tied High-Order Interactions

Although $f_{\boldsymbol{\beta}}$ is scalar, the term $\mathcal{S}^{\odot p}$ is not merely an independent output calibration. We arrive at it by first defining an explicit p-fold Tucker extension: instead of mapping one latent index per mode, the mode-n projection is a higher-order tensor $\mathcal{A}^{(n,p)} \in \mathbb{R}^{I_n \times R_n^p}$ that couples p latent indices. The case $p = 1$ is ordinary Tucker, while $p = 2$ gives a quadratic Tucker-style interaction between two latent Tucker tuples. Appendix A.1 gives the formal construction and tensor-network view.

The explicit construction is a reference model, not the parameterization we train. Its projection tensors grow as $I_n R_n^p$, and each degree would add a new set of mode projections. NTD-PL keeps the same high-order interaction pattern by tying the p-fold projection to the original Tucker factor:

$$\mathcal{A}^{(n,p)}_{i_n, r_n^{(1)}, \ldots, r_n^{(p)}} = \prod_{q=1}^{p} A^{(n)}_{i_n, r_n^{(q)}}.$$

Under this tying, the degree-p channel is exactly $S_{\mathbf{i}}^{p}$. Thus it still couples p latent Tucker index tuples, but all copies share the same mode factors. Across degrees, the only new free parameters are the scalar coefficients $\{\beta_p\}$, which mix these tied channels.

The latent tensor $\mathcal{S}$ and the coefficients $\boldsymbol{\beta}$ are fitted jointly, so the low-rank backbone is optimized together with the polynomial channels. Section 4 analyzes the induced rank expansion and the resulting separation from standard Tucker.

3.3 Masked Least-Squares Objective

Let $\Omega \in \{0,1\}^{I_1\times\cdots\times I_N}$ denote the observed-entry mask and let $|\Omega| = \sum_{\mathbf{i}} \Omega_{\mathbf{i}}$. We estimate the core, factors, and link coefficients by the regularized masked objective

$$\min_{\mathcal{X},\{\mathbf{A}^{(n)}\},\boldsymbol{\beta}} \frac{1}{2|\Omega|} \left\|\Omega \odot (\mathcal{Y} - f_{\boldsymbol{\beta}}(\mathcal{S}))\right\|_F^2 + \frac{\lambda_X}{2}\|\mathcal{X}\|_F^2 + \sum_{n=1}^{N} \frac{\lambda_n}{2}\left\|\mathbf{A}^{(n)}\right\|_F^2 + \frac{\lambda_\beta}{2}\|\boldsymbol{\beta}\|_2^2. \tag{4}$$

The same formulation covers full reconstruction by taking $\Omega \equiv \mathbf{1}$ and tensor completion by restricting the loss to the observed entries. The quadratic penalties regularize the Tucker backbone and the low-dimensional polynomial link.

4 Theory

The theory addresses the rank-lift view raised by the model definition. A scalar link has few free parameters, but its powers can induce ordinary Tucker tensors with larger multilinear ranks. We first make this induced structure explicit, then give a separation from matched-rank Tucker and an approximation route for nonlinear responses.

4.1 Rank Expansion and Separation

Proposition 1 (Shared-backbone rank expansion)**.** *Let $\mathcal{S} = [\![\mathcal{X}; \mathbf{A}^{(1)}, \ldots, \mathbf{A}^{(N)}]\!]$ have Tucker rank at most $(R_1, \ldots, R_N)$. For every integer $p \geq 1$, $\mathcal{S}^{\odot p}$ admits a Tucker representation with mode ranks at most*

$$\left(\binom{R_1+p-1}{p}, \ldots, \binom{R_N+p-1}{p}\right). \tag{5}$$

More explicitly, the ordinary Tucker factors of $\mathcal{S}^{\odot p}$ can be chosen as Veronese lifts of the original factors. For $\mathbf{A} \in \mathbb{R}^{I\times R}$, define $\mathbf{A}^{\langle p\rangle} \in \mathbb{R}^{I\times D}$, $D = \binom{R+p-1}{p}$, by

$$A_{i,\alpha}^{\langle p\rangle} = \prod_{r=1}^{R} A_{i,r}^{\alpha_r}, \tag{6}$$

where $\alpha \in \mathbb{N}^R$ ranges over $|\alpha| = p$. Then $\mathcal{S}^{\odot p}$ factors through $\mathbf{A}^{(1)\langle p\rangle}, \ldots, \mathbf{A}^{(N)\langle p\rangle}$.

This proposition makes the tying in Section 3.2 explicit in standard Tucker language: a degree-p channel is an ordinary Tucker tensor whose mode factors are polynomial feature lifts of the same backbone factors. The link therefore changes rank structure through tied features, not through an independent high-order parameter block.

Theorem 1 (Tensorial separation from standard Tucker)**.** *Let $N \geq 3$, $R \geq 2$, $p \geq 2$, and suppose $I_n \geq D$ for every n, where $D = \binom{R+p-1}{p}$. Then there exists an order-N tensor $\mathcal{Y} \in \mathbb{R}^{I_1\times\cdots\times I_N}$ that is representable by a degree-p NTD-PL model with Tucker backbone rank $(R, \ldots, R)$, but whose exact standard Tucker rank is*

$$(D, \ldots, D). \tag{7}$$

Equivalently, a degree-p polynomial link can generate tensors whose multilinear ranks are $(\binom{R+p-1}{p}, \ldots, \binom{R+p-1}{p})$ from the same Tucker backbone of rank $(R, \ldots, R)$.

The proof in Appendix B uses a CP-diagonal Tucker backbone and generic full-rank Veronese feature matrices. The theorem is an existence separation, not a universality claim over all high-rank Tucker tensors: it identifies a structured high-rank subset generated by a low-rank backbone.

4.2 Approximation Under Nonlinear Measurement Responses

Theorem 2 (Polynomial generation)**.** *Suppose $\mathcal{Y} = g(\mathcal{S}^\star)$, where $\mathcal{S}^\star$ has Tucker rank at most $(R_1, \ldots, R_N)$ and g is a polynomial of degree at most P. Then degree-P NTD-PL with the same Tucker rank can represent $\mathcal{Y}$ exactly.*

Theorem 3 (Analytic link approximation)**.** *Suppose $\mathcal{Y} = g(\mathcal{S}^\star)$, where $\mathcal{S}^\star$ has Tucker rank at most $(R_1, \ldots, R_N)$ and $\|\mathcal{S}^\star\|_\infty \leq B$. If g is analytic on the complex disk $D(0, \rho)$ with $B < \rho$, and $M_\rho = \sup_{|z|=\rho} |g(z)|$, then there exists a degree-P NTD-PL model with the same Tucker backbone satisfying*

$$\frac{1}{\sqrt{M}} \left\| \mathcal{Y} - \widehat{\mathcal{Y}} \right\|_F \leq M_\rho \frac{(B/\rho)^{P+1}}{1 - B/\rho}, \qquad M = \prod_{n=1}^{N} I_n. \tag{8}$$

Thus the same finite link that induces a tied rank lift also gives a direct approximation route for smooth responses on a bounded latent range. The statement uses the simple Taylor truncation bound; sharper polynomial approximation bounds can replace it under stronger assumptions.

Together, the results clarify the scope of NTD-PL. Polynomial links can approximate nonlinear responses while keeping the same latent Tucker rank, and their powers induce Veronese-lifted Tucker factors. This explains how a scalar link can change multilinear rank structure without introducing untied high-order tensor parameters.

5 Optimization

We optimize the masked objective in Section 3.3 by alternating between the scalar link and the Tucker backbone. The nonlinear part is low-dimensional once the latent Tucker tensor is fixed, while the backbone update remains a standard differentiable Tucker fitting step through the current link. We call the two stages *Link Refresh* and *Linked-Backbone Descent*.

Link Refresh. Fix the current latent tensor

$$S = [\![\mathcal{X}; \mathbf{A}^{(1)}, \ldots, \mathbf{A}^{(N)}]\!]. \tag{9}$$

For active degree $Q \leq P$, updating the polynomial coefficients reduces to a small masked ridge regression,

$$\min_{\boldsymbol{\beta}_{0:Q}} \frac{1}{2|\Omega|} \left\| y_\Omega - \mathbf{\Phi}_{\Omega,Q}(S) \boldsymbol{\beta}_{0:Q} \right\|_2^2 + \frac{\lambda_\beta}{2} \left\| \boldsymbol{\beta}_{0:Q} \right\|_2^2, \tag{10}$$

where $\mathbf{\Phi}_{\Omega,Q}(S)$ contains the observed powers $[1, s, \ldots, s^Q]$. This step is cheap because it solves only a $(Q+1) \times (Q+1)$ linear system. It also gives an interpretable view of the link: once the backbone exposes scalar latent values over the tensor, the refresh estimates the best scalar response on the observed set.

Linked-Backbone Descent. With $\boldsymbol{\beta}$ fixed, the Tucker parameters are updated through the current nonlinear measurement map. The masked residual is first propagated to the latent tensor as

$$T = \frac{1}{|\Omega|} \Omega \odot (f_{\boldsymbol{\beta}}(S) - Y) \odot f'_{\boldsymbol{\beta}}(S), \tag{11}$$

and then backpropagated through the Tucker reconstruction to update $\mathcal{X}$ and $\{\mathbf{A}^{(n)}\}$. In implementation we use Adam for these first-order steps, but the update itself is just standard Tucker reverse contractions composed with the current link. The backbone is therefore not calibrated after fitting; it is jointly reshaped under the nonlinear link.

Stabilization. Directly activating all polynomial degrees from the start is brittle, because higher powers of S amplify scale ambiguity. We therefore begin from the Tucker-compatible degree-one model and activate higher degrees gradually. When a new degree is switched on, its coefficient is initialized at zero, so the model expands from the current lower-degree solution without changing the prediction abruptly. We also normalize Tucker factor scales and absorb the corresponding scales into the core, which fixes a Tucker gauge without changing S. For the coefficient solve, the implementation may regress on a centered and scaled latent variable and then map the solution back to the original power basis. Appendix Figure 6 shows representative training curves where activating the relevant higher degrees produces systematic loss drops. Full gradients, stabilization details, costs analysis and pseudocode are given in Appendix A.2.

Masked uniform convergence. The masked objective also admits a standard sampling interpretation. Let $m = |\Omega|$ and let $\Theta \subset \mathbb{R}^d$ be a bounded NTD-PL parameter set, where $d = \prod_{n=1}^N R_n + \sum_{n=1}^N I_n R_n + (P+1)$. If the per-sample losses are bounded by $0 \le \ell_{\mathbf{i}}(\theta) \le B_\ell$ and G-Lipschitz in θ, then with probability at least $1-\delta$,

$$\sup_{\theta\in\Theta} \left| \mathcal{L}(\theta) - \widehat{\mathcal{L}}_m(\theta) \right| \le 2G\varepsilon + B_\ell \sqrt{\frac{d\log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}, \tag{12}$$

for every $0 < \varepsilon \le B_\Theta$. This is a standard uniform convergence bound, not a sharp minimax result, but it clarifies that masked training controls the full-tensor risk with a complexity term determined by the tied NTD-PL parameter count. The proof is given in Appendix B.4.

6 Experiments

The experiments test three mechanism questions: whether NTD-PL behaves as a parameter-tied rank lift under controlled nonlinear residuals; whether the polynomial link improves matched-rank Tucker on real reconstruction and held-out completion; and whether D_{link} diagnoses beneficial tensor units across domains. Unless stated otherwise, real tensors are globally max-normalized and used without spatial downsampling or cropping. We report RMSE and SAM for reconstruction, and RMSE^* and SAM^* only on held-out entries for completion. NTD-PL is compared mainly against same-rank Tucker, except in the rank-lift ablation.

6.1 Controlled Nonlinear Residuals

These synthetic experiments are mechanism checks rather than generic benchmarks. We generate measured tensors as

$$Y_\alpha = \sqrt{1-\alpha}S^\star + \sqrt{\alpha}\widetilde{R}^\star,$$

where $S^\star$ is a low-rank Tucker tensor and $\widetilde{R}^\star$ is the normalized component of a scalar nonlinear response orthogonal to $S^\star$. Thus α controls nonlinear residual energy, not just the raw amplitude of a nonlinearity. Figure 1 shows the expected transition: NTD-PL is Tucker-like when the residual is nearly linear and separates as nonlinear residual energy increases. Tucker can lag behind CP, but NTD-PL then pulls ahead systematically. Figure 2 shows the complementary degree effect: gains emerge when the active polynomial degree can match or approximate the response complexity. Degree continuation and coefficient-contribution details are deferred to Appendix C.1, where Table 4 shows the linear consistency negative control. Taken together, these checks indicate that NTD-PL is not a blanket improvement rule; its gain depends on the data's nonlinear structure.

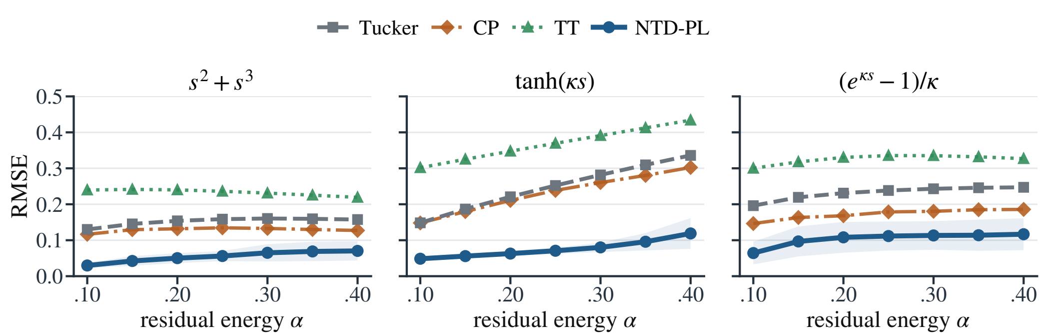

Figure 1: Controlled nonlinear residual sweep. NTD-PL stays Tucker-like near the linear regime and separates as α, the nonlinear residual energy, increases. Lower RMSE is better.

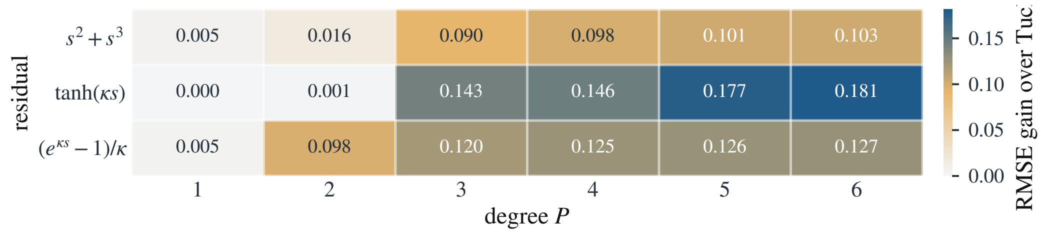

Figure 2: Degree-dependent gains at $\alpha = 0.25$. Cell values are Tucker minus NTD-PL RMSE; gains appear when the active degree is sufficient for the response complexity.

6.2 Tucker Enhancement on Real Hyperspectral Tensors

This section tests whether the same fixed-rank enhancement appears on real hyperspectral tensors, both when the tensor is fully measured and when evaluation is performed on held-out missing entries. We use CAVE dataset because it offers multiple real scenes with rich spectral and spatial structure, making it a reasonable mechanism check rather than a toy-sized tensor.

Representation enhancement. Full reconstruction is a representation-capacity check, not a generalization test. Figure 3 shows that NTD-PL improves RMSE and SAM on CAVE at matched Tucker backbone rank and similar parameter budget. The gap shrinks as Tucker rank increases, consistent with a fixed-rank enhancement: ordinary multilinear capacity already explains more of the tensor. Table 1 gives the complementary rank-lift ablation. Tucker can match the NTD-PL RMSE only after spending substantially more ordinary rank and parameters, supporting the parameter-tied rank-lift interpretation. Even when RMSE is close, NTD-PL usually retains a clearer SAM advantage.

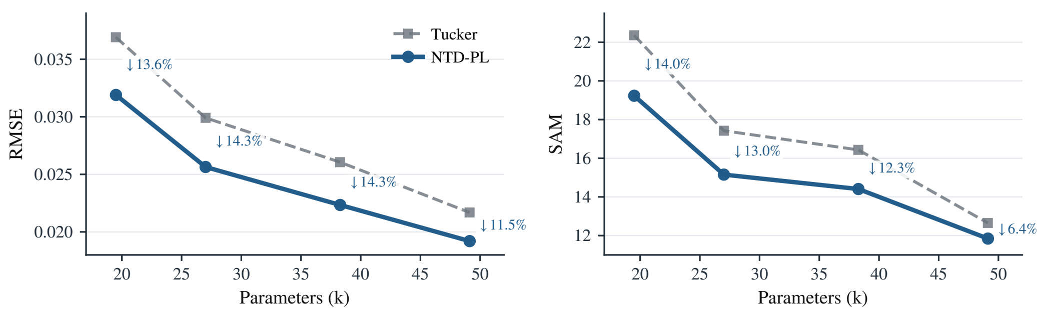

Figure 3: CAVE full reconstruction as a representation-capacity check. At matched Tucker backbone rank, NTD-PL reduces RMSE and SAM; annotations show paired relative gains over Tucker at each rank. The gap shrinks as Tucker rank increases, consistent with fixed-rank enhancement. Each point averages 15 scenes.

Table 1: Rank-lifted Tucker comparison on CAVE reconstruction. For each NTD-PL backbone rank, we increase Tucker's spatial ranks while keeping the spectral rank fixed and report the first Tucker rank that matches NTD-PL RMSE, along with the extra Tucker parameters required.

NTD-PL reference				First Tucker matching RMSE				Extra params
Rank	Params	RMSE↓	SAM↓	Rank	Params	RMSE↓	SAM↓	
$(18, 18, 3)$	20k	0.0319	**19.23**	$(28, 28, 3)$	31k	**0.0316**	21.41	$+59.5\%$
$(24, 24, 4)$	27k	0.0256	**15.15**	$(35, 35, 4)$	41k	**0.0254**	16.26	$+51.3\%$
$(33, 33, 4)$	38k	0.0223	**14.41**	$(49, 49, 4)$	60k	**0.0222**	15.28	$+56.5\%$

Held-out enhancement. Held-out completion is the stronger evidence because it tests whether the learned geometry transfers to unseen entries. Table 2 reports CAVE completion under random masks, with RMSE* and SAM* computed only on missing entries. At the same rank budget, NTD-PL gives consistent held-out RMSE* gains and usually improves SAM* as well. The paired sign and Wilcoxon tests in Appendix C.2 show the same scene-level pattern.

Table 2: Held-out CAVE completion at rank $(33, 33, 4)$ under random masks. RMSE* and SAM* are computed only on missing entries. Columns report mean $\pm$ standard deviation over scenes; gains are paired relative improvements over matched-rank Tucker.

Missing rate	RMSE*↓			SAM*↓		
	Tucker	NTD-PL	Gain	Tucker	NTD-PL	Gain
0.1	0.0263 ± 0.0179	$\mathbf{0.0229 \pm 0.0158}$	12.9%	11.82 ± 5.33	$\mathbf{8.36 \pm 3.38}$	29.3%
0.3	0.0263 ± 0.0179	$\mathbf{0.0230 \pm 0.0158}$	12.5%	14.75 ± 6.33	$\mathbf{12.47 \pm 5.06}$	15.5%
0.5	0.0264 ± 0.0179	$\mathbf{0.0230 \pm 0.0158}$	12.9%	15.56 ± 6.85	$\mathbf{13.48 \pm 5.75}$	13.4%
0.7	0.0266 ± 0.0180	$\mathbf{0.0219 \pm 0.0128}$	17.7%	16.02 ± 7.21	$\mathbf{13.79 \pm 5.43}$	13.9%

6.3 Diagnosing Nonlinear Low-Rank Structure Across Data

NTD-PL is meant to be diagnosable rather than uniformly beneficial. Its gain should appear when matched-rank Tucker leaves a coherent residual that a scalar link can reduce. We measure this with D_{link}, a cheap frozen-backbone Link Refresh probe: after fitting Tucker, freeze the backbone, run one scalar Link Refresh ridge solve, and measure residual-energy reduction on the relevant evaluation samples. Larger D_{link} indicates more link-like residual structure:

$$D_{\text{link}} = 10 \log_{10} \frac{\|Y_{\text{eval}} - \widehat{Y}_T\|_F^2}{\|Y_{\text{eval}} - \widehat{Y}_{\text{link}}\|_F^2},$$

where $\widehat{Y}_T$ is the Tucker output and $\widehat{Y}_{\text{link}}$ is the output after the one-step refresh.

For the CAVE completion, Table 3 groups scenes by D_{link} tercile. Higher D_{link} consistently corresponds to larger held-out RMSE* gains across missing rates, with Spearman correlations around 0.829–0.889. Appendix Figure 7 shows the corresponding scene-level scatter plots.

Table 3: CAVE completion diagnostic. Scenes are grouped by D_{link} terciles, and entries report mean NTD-PL RMSE* gains over matched-rank Tucker. Larger D_{link} consistently corresponds to larger held-out gains across missing rates.

Missing rate	Low D_{link}	Mid D_{link}	High D_{link}	Spearman ρ_s
0.1	0.75%	7.92%	12.63%	0.829
0.3	-1.13%	9.10%	13.12%	0.871
0.5	-1.09%	9.16%	13.18%	0.889
0.7	-0.62%	9.22%	13.07%	0.868

We then apply the same per-unit protocol beyond HSI. Each natural image is one tensor unit; each object or instance forms an object-view tensor; each KTH segment or UCF101 clip is an action-video tensor. Figure 4 plots D_{link} against NTD-PL RMSE gain across these independently fitted units. Diagnostic labels and terciles are assigned from D_{link}, not from NTD-PL gain, and the relationship is domain-dependent but positive: natural images show weaker link yield, object-view tensors clearer yield, and action-video clips an intermediate pattern.

These results support the scoped claim that NTD-PL helps when matched-rank Tucker leaves link-like residual structure, not uniformly across all visual tensors. Appendix Figure 7 gives the CAVE scene-level scatter supplement, and Appendix Table 7 gives the exact non-HSI numeric statistics.

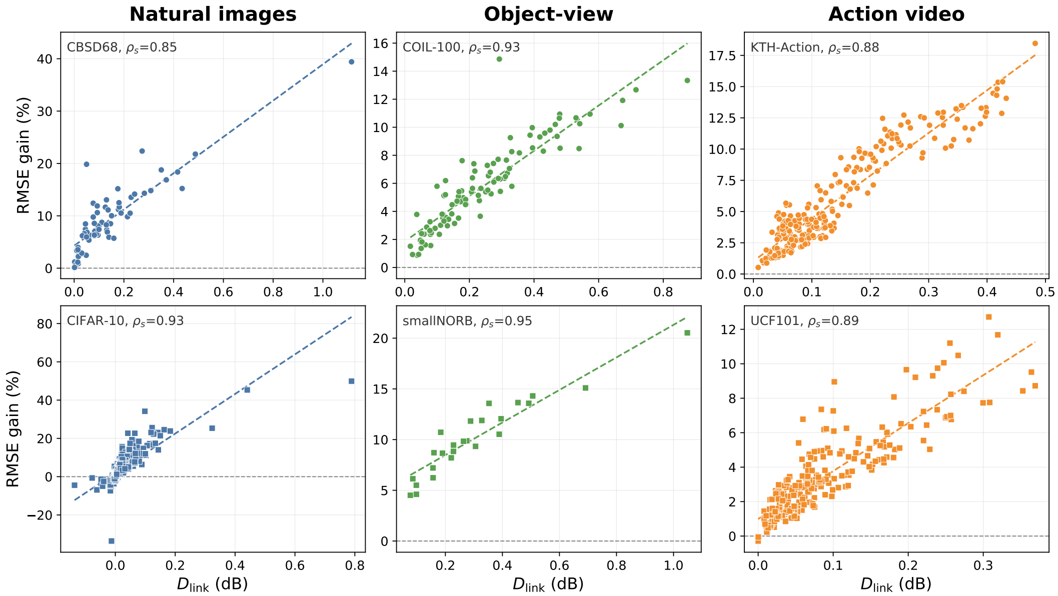

Figure 4: Per-unit non-HSI diagnostic correlation. Each panel plots D_{link} against NTD-PL RMSE gain over independently fitted tensor units. D_{link} is computed by frozen-backbone Link Refresh, and diagnostic labels are assigned from D_{link}, not from NTD-PL gain. The figure shows domain-dependent but positive diagnostic structure.

7 Limitations

NTD-PL assumes that a common nonlinear response explains a coherent part of the residual left by matched-rank Tucker. This scalar link may be less effective when nonlinearities vary strongly across spatial regions, spectral bands, modes, or samples. In such cases, a single polynomial response can underfit the domain variation even if the Tucker backbone is appropriate.

NTD-PL enhances a fixed Tucker rank budget by changing the measurement map on top of the latent tensor. When the remaining error is ordinary multilinear structure rather than a scalar response, increasing Tucker rank remains the more direct modeling choice.

Polynomial degree introduces a second tradeoff. Higher degrees increase the approximation capacity of the link, but they can amplify Tucker scale ambiguity and numerical conditioning issues in the power basis. This motivates the ridge-regularized Link Refresh step, factor normalization, and degree continuation used in the optimizer.

Finally, the joint problem remains nonconvex. The Link Refresh subproblem is a small ridge regression once the latent tensor is fixed, but the Tucker backbone and polynomial link are learned jointly through a nonconvex objective. We do not claim global optimality.

Our strongest empirical evidence is currently from controlled nonlinear residuals and hyperspectral reconstruction and completion. Broader validation on large sparse tensors, temporal or multimodal tensors, and domain-varying nonlinear responses is left for future work.

8 Conclusion

This paper develops polynomial links as parameter-tied rank lifts for Tucker tensors. The model keeps a Tucker latent tensor as the backbone and learns a polynomial link, adding only scalar coefficients. The resulting powers of the same latent tensor act as shared-backbone high-order interactions rather than untied high-rank corrections.

The analysis makes this rank-lift view explicit through Veronese rank expansion, separation from matched-rank Tucker, and approximation results for nonlinear links. The optimization uses a masked objective with Link Refresh, Linked-Backbone Descent, and degree continuation, while the experiments and diagnostics identify the intended regime: the link helps when matched-rank Tucker leaves a coherent nonlinear residual explained by a scalar response.

Future work may consider spatially or spectrally adaptive links, scalable sparse implementations, and richer structured links beyond scalar polynomials.

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A Additional Model Details

A.1 Explicit p-Fold Tucker Motivation

This section expands the motivation behind the explicit p-fold Tucker channel in Section 3.2. The construction is best viewed as the tensor analogue of adding polynomial interactions to a linear map. In the matrix case, one may move from a linear model $\mathbf{y} = \mathbf{A}\mathbf{x}$ to a quadratic model

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathcal{A}(\mathbf{x}, \mathbf{x}),$$

where the second term couples pairs of latent coordinates. The corresponding question for Tucker is: if standard Tucker is the multilinear low-rank map for multiway data, what is the direct p-th order interaction analogue?

For a Tucker backbone, an explicit p-fold interaction can be written as

$$\widetilde{Y}^{(p)}_{\mathbf{i}} = \sum_{\mathbf{r}^{(1)},\ldots,\mathbf{r}^{(p)}} \left(\prod_{q=1}^{p} \mathcal{X}_{r_1^{(q)},\ldots,r_N^{(q)}} \right) \prod_{n=1}^{N} \mathcal{A}^{(n,p)}_{i_n, r_n^{(1)},\ldots,r_n^{(p)}}. \tag{13}$$

Here each mode projection depends on p latent indices instead of one. For $p = 1$, this reduces to the ordinary Tucker entry formula. For $p = 2$, the same core appears twice and the mode projections couple two latent tuples, giving a quadratic Tucker-style channel. Figure 5 illustrates this progression.

The explicit construction is useful as a reference model, but it is not the parameterization used by NTD-PL. Its mode projections $\mathcal{A}^{(n,p)}$ have size $I_n R_n^p$, so a direct implementation introduces new high-order projection tensors and rapidly increases storage and contraction cost. NTD-PL keeps the same high-order interaction pattern by tying the projection tensor to the original Tucker factor:

$$\mathcal{A}^{(n,p)}_{i_n, r_n^{(1)},\ldots,r_n^{(p)}} = \prod_{q=1}^{p} A^{(n)}_{i_n, r_n^{(q)}}. \tag{14}$$

Substituting (14) into (13) gives exactly $\mathcal{S}^{\odot p}$, where $\mathcal{S} = [\![\mathcal{X}; \mathbf{A}^{(1)}, \ldots, \mathbf{A}^{(N)}]\!]$. Thus the degree-$p$ NTD-PL channel is a shared-projection version of the explicit p-fold Tucker prototype.

This is the modeling role of the p-fold Tucker prototype: it defines the high-order Tucker interaction that NTD-PL compresses. The final model does not train the untied tensors $\mathcal{A}^{(n,p)}$; it keeps a single Tucker backbone and adds only the polynomial coefficients $\{\beta_p\}$.

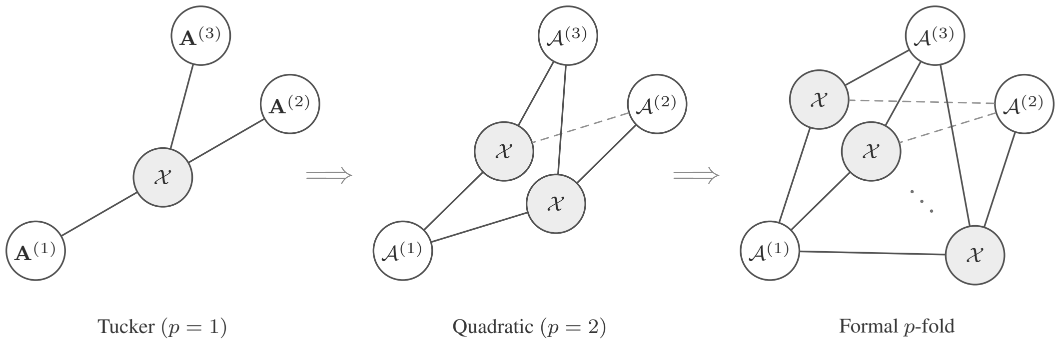

Figure 5: Tensor-network view of a formal p-fold Tucker extension. The construction keeps the Tucker idea of mode-wise generation, but lets each mode projection couple multiple latent index tuples.

A.2 Optimization Details

This section gives the update equations behind the solver summarized in Section 5. The implementation follows the same structure: one Tucker reconstruction per iteration, a fused evaluation of the polynomial and its derivative, first-order updates of the backbone, and periodic ridge-regression updates of the link coefficients.

Backbone gradients. For a fixed $\boldsymbol{\beta}$, define

$$f'_{\boldsymbol{\beta}}(\mathcal{S}) = \sum_{p=1}^{P} p\beta_p \mathcal{S}^{\odot(p-1)}. \tag{15}$$

The gradient of the normalized masked data term with respect to the latent tensor $\mathcal{S}$ is

$$\mathbf{T} = \frac{1}{|\Omega|}\Omega \odot (f_{\boldsymbol{\beta}}(\mathcal{S}) - \mathcal{Y}) \odot f'_{\boldsymbol{\beta}}(\mathcal{S}). \tag{16}$$

Backpropagating $\mathbf{T}$ through the Tucker reconstruction gives the core gradient

$$\nabla_{\mathcal{X}}\mathcal{L} = \mathbf{T} \times_1 \mathbf{A}^{(1)\top} \times_2 \cdots \times_N \mathbf{A}^{(N)\top} + \lambda_X \mathcal{X}. \tag{17}$$

For the n-th factor, let

$$\mathbf{Z}^{(n)} = \left[\mathcal{X} \times_1 \mathbf{A}^{(1)} \cdots \times_{n-1} \mathbf{A}^{(n-1)} \times_{n+1} \mathbf{A}^{(n+1)} \cdots \times_N \mathbf{A}^{(N)}\right]_{(n)}. \tag{18}$$

Then

$$\nabla_{\mathbf{A}^{(n)}}\mathcal{L} = \mathbf{T}_{(n)}\mathbf{Z}^{(n)\top} + \lambda_n \mathbf{A}^{(n)}. \tag{19}$$

The experiments use Adam for these first-order steps. This choice is not part of the model definition; any comparable first-order optimizer can use the same gradients.

Link update. With $\mathcal{S}$ fixed and active degree $Q = P_{\text{act}}$, the coefficient subproblem is the ridge regression in (10). Its closed-form solution is

$$\boldsymbol{\beta}^{\star}_{0:Q} = \left(\boldsymbol{\Phi}^{\top}_{\Omega,Q}\boldsymbol{\Phi}_{\Omega,Q} + |\Omega|\lambda_{\beta}\mathbf{I}\right)^{-1} \boldsymbol{\Phi}^{\top}_{\Omega,Q}\mathbf{y}_{\Omega}. \tag{20}$$

Equivalently, one may absorb $|\Omega|$ into the ridge parameter and solve the usual unnormalized normal equations. If the constant term is disabled, the first column is removed and β_0 is fixed at zero. In the code, this small system is solved either from an explicit Vandermonde design or from moment equations built from cached powers.

Stabilization and continuation. High powers of $\mathcal{S}$ can be sensitive to scale. The implementation therefore normalizes factor columns after backbone updates and absorbs the corresponding scales into the core; this is a gauge fixing of the Tucker parameterization and does not change $\mathcal{S}$. For the coefficient solve, it may regress on the centered and scaled latent variable $(S - \mu)/\sigma$, then convert the solution back to the original power basis of S.

The active degree follows the schedule

$$t_k = \mathrm{round}\left(\frac{kT}{P}\right), \qquad k = 1, \dots, P-1.$$

When degree $p+1$ is activated, the previous coefficients and Tucker factors are kept and the new coefficient β_{p+1} is initialized at zero.

Cost. The dominant cost is forming the Tucker reconstruction and its reverse contractions on the observed tensor. For dense tensors this is comparable to standard Tucker optimization at the same rank, plus $O(P|\Omega|)$ work to evaluate the scalar polynomial and its derivative. The link update forms a Vandermonde design or the corresponding moment equations and solves a $(P+1)\times(P+1)$ linear system, with direct cost $O(P^2|\Omega| + P^3)$. Since P is small, this cost is negligible compared with the Tucker contractions.

Algorithm 1 gives the full procedural summary.

Algorithm 1 Masked NTD-PL optimization

Require: observed tensor $\mathcal{Y}$, mask Ω, rank $(R_1, \dots, R_N)$, degree P

initialize $\mathcal{X}, \{\mathbf{A}^{(n)}\}$ from Tucker on the masked tensor
initialize active degree $P_{\text{act}} \leftarrow 1$ and $\boldsymbol{\beta} = (0, 1)$
for $t = 1, \dots, T$ **do**
    form $\mathcal{S} = [\![\mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)}]\!]$
    if a link update is scheduled **then**
        update active coefficients $\boldsymbol{\beta}_{0:P_{\text{act}}}$ by ridge regression on $\{(S_{\mathbf{i}}, Y_{\mathbf{i}}) : \Omega_{\mathbf{i}} = 1\}$
    end if
    compute $f_{\boldsymbol{\beta}}(\mathcal{S})$ and $f'_{\boldsymbol{\beta}}(\mathcal{S})$
    take first-order steps on $\mathcal{X}$ and $\{\mathbf{A}^{(n)}\}$ using the masked gradient
    normalize factor scales and absorb them into the core
    increase P_{act} according to the continuation schedule, initializing the new coefficient at zero
end for
return $\mathcal{X}, \{\mathbf{A}^{(n)}\}, \boldsymbol{\beta}$

B Additional Proofs

B.1 Polynomial and Analytic Approximation

Proof of Theorem 2. Since g has degree at most P, write

$$g(s) = \sum_{p=0}^{P} c_p s^p. \tag{21}$$

Take the NTD-PL latent tensor to be the true latent tensor, $\mathcal{S} = \mathcal{S}^\star$, and set $\beta_p = c_p$. Then, entrywise,

$$\widehat{\mathcal{Y}}_{i_1,\dots,i_N} = \sum_{p=0}^{P} \beta_p \left(\mathcal{S}^\star_{i_1,\dots,i_N}\right)^p = g\left(\mathcal{S}^\star_{i_1,\dots,i_N}\right) = \mathcal{Y}_{i_1,\dots,i_N}. \tag{22}$$

Thus $\widehat{\mathcal{Y}} = \mathcal{Y}$. □

Lemma 1 (Scalar-to-tensor approximation)**.** *Let $\|\mathcal{S}^\star\|_\infty \le B$. For any scalar function g and any degree-P polynomial q_P,*

$$\|g(\mathcal{S}^\star) - q_P(\mathcal{S}^\star)\|_F \le \sqrt{\prod_{n=1}^{N} I_n}\, \sup_{|s|\le B} |g(s) - q_P(s)|. \tag{23}$$

Proof. Every entry of $\mathcal{S}^\star$ lies in $[-B, B]$, so every entrywise approximation error is bounded by $\sup_{|s|\le B} |g(s) - q_P(s)|$. Squaring, summing over all entries, and taking the square root gives the claim. □

Proof of Theorem 3. Let $g(s) = \sum_{k=0}^{\infty} a_k s^k$ be the Taylor expansion of g around zero, and choose $q_P(s) = \sum_{k=0}^{P} a_k s^k$. By Cauchy's estimate,

$$|a_k| \leq \frac{M_\rho}{\rho^k}. \tag{24}$$

For $|s| \leq B < \rho$,

$$|g(s) - q_P(s)| \leq \sum_{k=P+1}^{\infty} |a_k|\,|s|^k \tag{25}$$

$$\leq M_\rho \sum_{k=P+1}^{\infty} (B/\rho)^k = M_\rho \frac{(B/\rho)^{P+1}}{1 - B/\rho}. \tag{26}$$

Applying Lemma 1 and taking $\mathcal{S} = \mathcal{S}^\star$, $\beta_k = a_k$ for $0 \leq k \leq P$, yields the stated unnormalized bound. Dividing by $\sqrt{M}$, $M = \prod_{n=1}^{N} I_n$, gives the normalized form used in Theorem 3. $\square$

B.2 Rank Expansion

Proof of Proposition 1. Write the Tucker tensor entrywise as

$$\mathcal{S}_{i_1,\ldots,i_N} = \sum_{r_1=1}^{R_1} \cdots \sum_{r_N=1}^{R_N} \mathcal{X}_{r_1,\ldots,r_N} \prod_{n=1}^{N} A^{(n)}_{i_n,r_n}. \tag{27}$$

Taking an element-wise p-th power gives

$$(\mathcal{S}_{i_1,\ldots,i_N})^p = \prod_{q=1}^{p} \left(\sum_{r_1^{(q)}=1}^{R_1} \cdots \sum_{r_N^{(q)}=1}^{R_N} \mathcal{X}_{r_1^{(q)},\ldots,r_N^{(q)}} \prod_{n=1}^{N} A^{(n)}_{i_n,r_n^{(q)}} \right) \tag{28}$$

$$= \sum_{\gamma_1 \in [R_1]^p} \cdots \sum_{\gamma_N \in [R_N]^p} \left(\prod_{q=1}^{p} \mathcal{X}_{r_1^{(q)},\ldots,r_N^{(q)}} \right) \prod_{n=1}^{N} \left(\prod_{q=1}^{p} A^{(n)}_{i_n,r_n^{(q)}} \right), \tag{29}$$

where $\gamma_n = (r_n^{(1)}, \ldots, r_n^{(p)})$. For each mode n, let $\alpha_n \in \mathbb{N}^{R_n}$ record the counts of the entries in γ_n, so $|\alpha_n| = p$. Then

$$\prod_{q=1}^{p} A^{(n)}_{i_n,r_n^{(q)}} = \prod_{r=1}^{R_n} \left(A^{(n)}_{i_n,r} \right)^{\alpha_{n,r}} = A^{(n)\langle p \rangle}_{i_n,\alpha_n}. \tag{30}$$

Group all ordered tuples $(\gamma_1, \ldots, \gamma_N)$ that produce the same multi-indices $(\alpha_1, \ldots, \alpha_N)$, and define the aggregated core

$$\mathcal{X}^{\langle p \rangle}_{\alpha_1,\ldots,\alpha_N} = \sum_{\substack{\gamma_1,\ldots,\gamma_N: \\ \mathrm{count}(\gamma_n)=\alpha_n \ \forall n}} \prod_{q=1}^{p} \mathcal{X}_{r_1^{(q)},\ldots,r_N^{(q)}}. \tag{31}$$

The preceding display is exactly the Tucker reconstruction

$$\mathcal{S}^{\odot p} = [\![\mathcal{X}^{\langle p \rangle}; \mathbf{A}^{(1)\langle p \rangle}, \ldots, \mathbf{A}^{(N)\langle p \rangle}]\!]. \tag{32}$$

The n-th Veronese factor has $\binom{R_n+p-1}{p}$ columns, giving the claimed Tucker rank bound. $\square$

B.3 Separation

Lemma 2 (Full-rank Veronese evaluations)**.** *Let $D = \binom{R+p-1}{p}$. There exists a matrix $\mathbf{A} \in \mathbb{R}^{D \times R}$ whose Veronese feature matrix $\mathbf{A}^{\langle p \rangle} \in \mathbb{R}^{D \times D}$ has rank D. Consequently, for $I \geq D$, a generic matrix $\mathbf{A} \in \mathbb{R}^{I \times R}$ has $\mathrm{rank}(\mathbf{A}^{\langle p \rangle}) = D$.*

Proof. The homogeneous degree-p monomials in R variables are linearly independent polynomials. Hence there exist evaluation points $x_1, \dots, x_D \in \mathbb{R}^R$ such that the matrix $(x_i^\alpha)_{i,\alpha}$, $|\alpha| = p$, is nonsingular; otherwise every $D \times D$ evaluation determinant would vanish, which would imply a nontrivial linear dependence among the monomials. Taking these points as the rows of $\mathbf{A}$ gives the first claim.

For $I \geq D$, choose any D rows. The determinant of the corresponding $D \times D$ Veronese submatrix is a polynomial in the entries of $\mathbf{A}$ that is not identically zero by the first claim. Its nonvanishing is therefore a generic property, so $\mathbf{A}^{\langle p \rangle}$ has full column rank for generic $\mathbf{A}$. $\square$

Proof of Theorem 1. Let $D = \binom{R+p-1}{p}$. Choose generic factor matrices $\mathbf{A}^{(n)} = [a_1^{(n)}, \dots, a_R^{(n)}] \in \mathbb{R}^{I_n \times R}$, and define the CP-diagonal Tucker backbone

$$\mathcal{S} = \sum_{r=1}^{R} a_r^{(1)} \otimes \cdots \otimes a_r^{(N)}. \tag{33}$$

Its mode-n unfolding is

$$\mathcal{S}_{(n)} = \mathbf{A}^{(n)} \left(\mathbf{A}^{(N)} \odot \cdots \odot \mathbf{A}^{(n+1)} \odot \mathbf{A}^{(n-1)} \odot \cdots \odot \mathbf{A}^{(1)} \right)^\top. \tag{34}$$

For generic $\mathbf{A}^{(n)}$, $\mathbf{A}^{(n)}$ has rank R, and the Khatri-Rao product on the right has rank R. Thus $\operatorname{rank}(\mathcal{S}_{(n)}) = R$ for every n, so $\mathcal{S}$ has exact Tucker rank $(R, \dots, R)$.

Set $\beta_p = 1$ and $\beta_q = 0$ for $q \neq p$. Then $\mathcal{Y} = \mathcal{S}^{\odot p}$ is a valid degree-p NTD-PL tensor. Entrywise,

$$\mathcal{Y}_{i_1,\dots,i_N} = \left(\sum_{r=1}^{R} \prod_{n=1}^{N} A_{i_n,r}^{(n)} \right)^p \tag{35}$$

$$= \sum_{|\alpha|=p} c_\alpha \prod_{n=1}^{N} \prod_{r=1}^{R} \left(A_{i_n,r}^{(n)} \right)^{\alpha_r}, \tag{36}$$

where $c_\alpha = p!/(\alpha_1! \cdots \alpha_R!) > 0$. Let $\mathbf{B}^{(n)} = \mathbf{A}^{(n)\langle p \rangle} \in \mathbb{R}^{I_n \times D}$. By Lemma 2, each $\mathbf{B}^{(n)}$ has full column rank D for generic $\mathbf{A}^{(n)}$. The mode-n unfolding is

$$\mathcal{Y}_{(n)} = \mathbf{B}^{(n)} \operatorname{diag}(c) \left(\mathbf{B}^{(N)} \odot \cdots \odot \mathbf{B}^{(n+1)} \odot \mathbf{B}^{(n-1)} \odot \cdots \odot \mathbf{B}^{(1)} \right)^\top. \tag{37}$$

The diagonal matrix $\operatorname{diag}(c)$ is nonsingular. Since a Khatri-Rao product of full-column-rank matrices is full column rank, the right factor has rank D. Therefore $\operatorname{rank}(\mathcal{Y}_{(n)}) = D$. This holds for every mode n, and the exact Tucker rank of $\mathcal{Y}$ is $(D, \dots, D)$. $\square$

B.4 Masked Generalization

Theorem 4 (Masked-entry uniform convergence)**.** *Let $\Theta \subset \mathbb{R}^d$ be a bounded NTD-PL parameter set with*

$$d = \prod_{n=1}^{N} R_n + \sum_{n=1}^{N} I_n R_n + (P+1).$$

For $\theta \in \Theta$, define the full-entry risk

$$\mathcal{L}(\theta) = \frac{1}{M} \sum_{\mathbf{i}} \ell_{\mathbf{i}}(\theta), \qquad M = \prod_{n=1}^{N} I_n,$$

and the masked empirical risk

$$\widehat{\mathcal{L}}_m(\theta) = \frac{1}{m} \sum_{t=1}^{m} \ell_{\mathbf{i}_t}(\theta),$$

where $\mathbf{i}_t$ are sampled uniformly from tensor entries. Assume $0 \le \ell_{\mathbf{i}}(\theta) \le B_\ell$ and $\ell_{\mathbf{i}}(\theta)$ is G-Lipschitz in θ, uniformly over $\mathbf{i}$. If Θ is contained in a Euclidean ball of radius B_Θ, then with probability at least $1-\delta$,

$$\sup_{\theta\in\Theta}\left|\mathcal{L}(\theta)-\widehat{\mathcal{L}}_m(\theta)\right| \le 2G\varepsilon + B_\ell\sqrt{\frac{d\log(3B_\Theta/\varepsilon)+\log(2/\delta)}{2m}}$$

for every $0<\varepsilon\le B_\Theta$.

Proof of Theorem 4. Fix $0<\varepsilon\le B_\Theta$, and let $\mathcal{N}_\varepsilon$ be an ε-net of Θ in Euclidean norm. Since Θ lies in a d-dimensional ball of radius B_Θ, it admits a net with

$$|\mathcal{N}_\varepsilon| \le \left(\frac{3B_\Theta}{\varepsilon}\right)^d. \tag{38}$$

For any fixed $\theta_0 \in \mathcal{N}_\varepsilon$, the random variables $\ell_{\mathbf{i}_t}(\theta_0)$ are independent and bounded in $[0, B_\ell]$. Hoeffding's inequality gives

$$\Pr\left(\left|\mathcal{L}(\theta_0)-\widehat{\mathcal{L}}_m(\theta_0)\right| \ge u\right) \le 2\exp\left(-\frac{2mu^2}{B_\ell^2}\right). \tag{39}$$

Taking a union bound over $\mathcal{N}_\varepsilon$, with probability at least $1-\delta$,

$$\max_{\theta_0\in\mathcal{N}_\varepsilon}\left|\mathcal{L}(\theta_0)-\widehat{\mathcal{L}}_m(\theta_0)\right| \le B_\ell\sqrt{\frac{\log(2|\mathcal{N}_\varepsilon|/\delta)}{2m}}. \tag{40}$$

Substituting the net-size bound yields the stated stochastic term.

Now take any $\theta\in\Theta$, and choose $\theta_0\in\mathcal{N}_\varepsilon$ with $\|\theta-\theta_0\|_2\le\varepsilon$. By the uniform Lipschitz assumption,

$$|\mathcal{L}(\theta)-\mathcal{L}(\theta_0)| \le G\varepsilon, \qquad |\widehat{\mathcal{L}}_m(\theta)-\widehat{\mathcal{L}}_m(\theta_0)| \le G\varepsilon. \tag{41}$$

Therefore

$$\left|\mathcal{L}(\theta)-\widehat{\mathcal{L}}_m(\theta)\right| \le |\mathcal{L}(\theta)-\mathcal{L}(\theta_0)| + \left|\mathcal{L}(\theta_0)-\widehat{\mathcal{L}}_m(\theta_0)\right| \tag{42}$$

$$+ |\widehat{\mathcal{L}}_m(\theta_0)-\widehat{\mathcal{L}}_m(\theta)| \tag{43}$$

$$\le 2G\varepsilon + B_\ell\sqrt{\frac{d\log(3B_\Theta/\varepsilon)+\log(2/\delta)}{2m}}. \tag{44}$$

Taking the supremum over $\theta\in\Theta$ completes the proof. □

Remark 1. *The Lipschitz and bounded-loss assumptions are standard for compact parametric classes. For NTD-PL, they follow when the observed values, core, factor matrices, and polynomial coefficients are bounded and the polynomial degree P is finite. The resulting Lipschitz constant may depend polynomially on these bounds and on P, but the covering dimension remains the number of tied NTD-PL parameters rather than the number of parameters in an untied high-order Tucker expansion.*

C Additional Experiments

C.1 Controlled Synthetic Mechanism Checks

The controlled experiments in the main text are designed to isolate the role of the scalar link. Table 4 first checks a negative control: when the source is linear Tucker, NTD-PL does not improve by inventing nonlinear terms. The only exception is an affine offset, where the constant link term is the intended correction.

For the nonlinear controlled tensors, the nonlinearity is introduced as an orthogonal residual rather than by directly setting $Y = g(S^\star)$. Let $S^\star$ be the linear low-rank Tucker backbone and let g be one of the candidate scalar responses. We form

$$R^\star = g(S^\star) - \frac{\langle g(S^\star), S^\star\rangle}{\|S^\star\|_F^2}\, S^\star, \tag{45}$$

and, when $\|R^\star\|_F > 0$, normalize it as

$$\widetilde{R}^\star = \frac{\|S^\star\|_F}{\|R^\star\|_F} R^\star. \tag{46}$$

The observed tensor is then

$$Y_\alpha = \sqrt{1-\alpha}\, S^\star + \sqrt{\alpha}\, \widetilde{R}^\star, \qquad 0 \le \alpha \le 1. \tag{47}$$

By construction, $\langle S^\star, \widetilde{R}^\star \rangle = 0$ and $\|\widetilde{R}^\star\|_F = \|S^\star\|_F$, so α is exactly the energy fraction assigned to the nonlinear residual. This keeps the total energy fixed while moving the observation away from the linear Tucker backbone. The candidate residual families are $g(s) = s^2 + s^3$, $g(s) = (\exp(\kappa s) - 1)/\kappa$, and $g(s) = \tanh(\kappa s)$, with κ set from the input range.

The main text reports the nonlinear alpha sweep and degree sweep. Figure 6 shows that degree continuation activates higher powers with systematic loss drops at relevant degrees. Table 5 complements those figures with a small but consistent term-mass concentration effect: the dominant contribution usually lands on the degree that matches the injected response, while smooth odd links spread mass across nearby odd orders.

Table 4: Linear consistency on synthetic Tucker tensors. In the linear negative control, effective $p > 1$ contributions remain small except for the intended affine offset case.

Method	bias $= 0$		bias $= 0.5$	
	RMSE↓	$p > 1$	RMSE↓	$p > 1$
Tucker	**0.0114**	–	0.0647	–
NTD-PL, $\beta_0 = 0$	0.0116	4.19%	0.0522	39.17%
NTD-PL, β_0 free	0.0117	3.19%	**0.0118**	2.98%

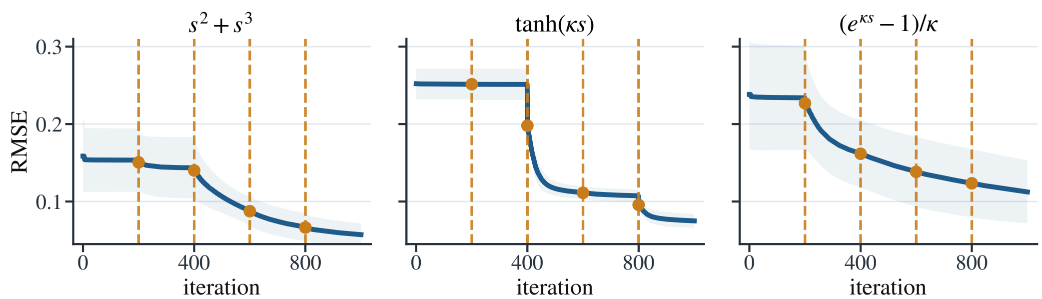

Figure 6: NTD-PL training curves for controlled nonlinear mechanism checks at $\alpha = 0.25$ and $P = 5$. Dashed vertical lines mark degree activations, which produce systematic drops when the activated degree matches the response structure.

Table 5: Effective NTD-PL term contributions under moderate nonlinearity, reported as percentages over ten seeds. Polynomial targets concentrate mass at matching nonlinear degrees, while odd smooth links place most nonlinear mass on odd powers.

Link	Effective contribution (%)						
	$p = 0$	$p = 1$	$p = 2$	$p = 3$	$p = 4$	$p = 5$	$p = 6$
$s^2 + s^3$	0.1	24.1	9.8	**38.1**	7.3	11.5	9.1
$(e^{\kappa s} - 1)/\kappa$	0.3	**39.8**	16.8	13.2	10.1	7.7	12.1
$\tanh(\kappa s)$	0.2	37.9	1.3	**30.9**	5.4	17.3	7.0

C.2 CAVE Completion Significance and Robustness

The main text reports the full-resolution random-missing CAVE completion table at rank $(33, 33, 4)$. We do not repeat that table here; instead, Table 6 gives the paired sign and Wilcoxon tests for the same protocol.

Table 6: Paired significance tests for full-resolution CAVE completion at rank $(33, 33, 4)$. Metrics are computed only on missing entries.

	RMSE				SAM			
ρ	W/L	Med. gain	Sign p	Wilcoxon p	W/L	Med. gain	Sign p	Wilcoxon p
0.1	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	14/1	2.9742	9.77×10^{-4}	8.54×10^{-4}
0.3	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	13/2	1.9194	0.007	0.004
0.5	14/1	0.0026	9.77×10^{-4}	1.22×10^{-4}	13/2	1.6566	0.007	0.012
0.7	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	12/3	1.3394	0.035	0.048

C.3 Link Refresh Residual Diagnostic

The diagnostic used in the main text reuses the Link Refresh step from Section 5 as a one-round residual probe. Let $\widehat{\mathcal{Y}}_{\mathrm{T}}$ denote the Tucker output and $\mathcal{Y}$ the target tensor. Starting from this fitted Tucker latent tensor, we update only the scalar polynomial coefficients by fitting

$$h_\gamma(x) = \sum_{p=0}^{4} \gamma_p x^p$$

by ridge regression on pairs $(\widehat{Y}_{\mathrm{T},\mathbf{i}}, Y_{\mathbf{i}})$. In full reconstruction, all entries are used for this fit. In completion, the fit uses only observed entries and is then evaluated on held-out entries. The fitted map is applied entrywise to $\widehat{\mathcal{Y}}_{\mathrm{T}}$. We summarize this cheap refresh by the Link Refresh diagnostic

$$D_{\mathrm{link}} = 10 \log_{10} \left(\frac{\|\mathcal{Y}_{\mathrm{eval}} - \widehat{\mathcal{Y}}_{\mathrm{T}}\|_F^2}{\|\mathcal{Y}_{\mathrm{eval}} - \widehat{\mathcal{Y}}_{\mathrm{link}}\|_F^2} \right),$$

which measures, on a log scale, how much Tucker residual energy is removed on the task-specific evaluation entries. In other words, it measures how much shared scalar-link yield is already present before the full NTD-PL alternating solver starts reshaping the Tucker parameters.

C.4 CAVE Completion Scene Diagnostics

Figure 7 supplements the main CAVE diagnostic table with the scene-level scatter plots used to visualize the link yield across missing rates. The plots show the same diagnostic relationship between D_{link} and held-out RMSE* gain, now at the scene level.

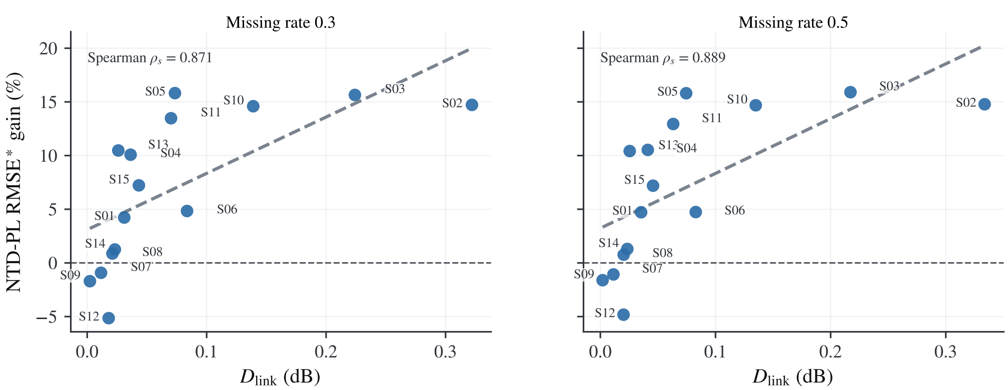

Figure 7: Scene-level supplement for the CAVE completion diagnostic table. Missing rates 0.3 and 0.5 are shown; D_{link} from frozen- backbone Link Refresh predicts larger held-out RMSE* gains.

C.5 Non-HSI Link Refresh Diagnostics

The main diagnostic table uses per-unit non-HSI experiments rather than collection-level tensors. CBSD68 uses per-image units resized to $96 \times 96 \times 3$ with rank $(32, 32, 3)$; the available diagnostic CSV contains 16 CBSD68 units. CIFAR-10 uses 1000 class-balanced test images of shape $32 \times 32 \times 3$ with rank $(20, 20, 3)$. COIL-100 uses one object at a time as a $36 \times 48 \times 48 \times 3$ view tensor with rank $(8, 12, 12, 2)$. smallNORB uses one object instance at a time as an $18 \times 64 \times 64 \times 2$ stereo view tensor with rank $(12, 24, 24, 2)$. All units are max-normalized independently, and diagnostic labels are assigned only from D_{link}, not from NTD-PL gain.

Table 7: Exact per-dataset statistics behind the main non-HSI scatter figure. Each unit is independently fitted by Tucker and NTD-PL at matched backbone rank. D_{link} is computed by frozen-backbone Link Refresh; diagnostic tiers are assigned from D_{link}, not from NTD-PL gain.

Missing rate	Low D_{link}	Mid D_{link}	High D_{link}	Spearman ρ_s
0.1	0.75%	7.92%	12.63%	0.829
0.3	-1.13%	9.10%	13.12%	0.871
0.5	-1.09%	9.16%	13.18%	0.889
0.7	-0.62%	9.22%	13.07%	0.868

NeurIPS Paper Checklist

This draft uses a working checklist. It should be replaced by the official NeurIPS checklist distributed with the final 2026 submission template.

1. **Claims.** The main claims are stated in the abstract, introduction, theory section, and experiment captions: NTD-PL is a Tucker enhancement and shared-backbone rank lift for Tucker tensors; its gains are tied to jointly reshaping the Tucker backbone under a shared scalar link rather than to modest rank increases alone.
2. **Limitations.** Section 7 discusses the shared scalar-link assumption, rank-selection scope, polynomial-degree stability, nonconvex optimization, and the current experimental scope.
3. **Theory assumptions.** The approximation results assume a bounded latent tensor and polynomial or analytic scalar links. The separation results use generic full-rank feature constructions. The masked generalization bound assumes bounded parameters, bounded observations, and random observed entries.
4. **Reproducibility.** The experiment section reports datasets, ranks, mask types, missing rates, and baseline definitions. The codebase includes scripts for rank sweeps, MLPCal, structured missingness, and the cross-domain real-tensor reconstruction study; the final version should add hardware details and exact command manifests.